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#include <Wire.h>
#include "MAX30105.h"
#include "heartRate.h"

MAX30105 particleSensor;

const byte RATE_SIZE = 10;
byte rates[RATE_SIZE];
byte rateSpot = 0;

long lastBeat = 0;

float beatsPerMinute;
int beatAvg;

unsigned long startTime;
bool measuring = false;

void setup()
{
  Serial.begin(115200);
  Serial.println("Initializing...");

  // Initialize sensor
  if (!particleSensor.begin(Wire, I2C_SPEED_STANDARD))
  {
    Serial.println("MAX30102 not found. Check wiring.");
    while (1);
  }

  Serial.println("Place your finger on the sensor...");

  particleSensor.setup();
  particleSensor.setPulseAmplitudeRed(0x0A);
  particleSensor.setPulseAmplitudeGreen(0);

  delay(1000);
}

void loop()
{
  long irValue = particleSensor.getIR();

  // Detect finger
  if (irValue > 50000)
  {
    if (!measuring)
    {
      Serial.println("Finger detected!");
      Serial.println("Measuring for 10 seconds...");

      measuring = true;
      startTime = millis();
      rateSpot = 0;
    }
  }
}

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    }

    // Detect heartbeat
    if (checkForBeat(irValue))
    {
        long delta = millis() - lastBeat;
        lastBeat = millis();

        beatsPerMinute = 60 / (delta / 1000.0);

        // Valid BPM range
        if (beatsPerMinute < 255 && beatsPerMinute > 20)
        {
            rates[rateSpot++] = (byte)beatsPerMinute;
            rateSpot %= RATE_SIZE;

            // Calculate average
            beatAvg = 0;
            for (byte x = 0; x < RATE_SIZE; x++)
                beatAvg += rates[x];

            beatAvg /= RATE_SIZE;

            Serial.print("Current BPM: ");
            Serial.println(beatsPerMinute);
        }
    }

    // After 10 seconds
    if (millis() - startTime >= 10000)
    {
        Serial.println("=====");
        Serial.print("AVERAGE BPM: ");
        Serial.println(beatAvg);
        Serial.println("Measurement complete.");
        Serial.println("Remove finger to restart.");
        Serial.println("=====");

        measuring = false;

        // Wait until finger removed
        while (particleSensor.getIR() > 50000)
        {
            delay(100);
        }

        Serial.println("Place finger again for new reading.");
    }
}
else
{
    Serial.println("No finger detected.");
    delay(500);
}
}

```

